

AIR UNIVERSITY

Department of Computer Science & Artificial Intelligence

FINAL YEAR PROJECT REPORT

Dengue Disease Detection System Using Machine Learning

A Comparative Study of Logistic Regression, Decision Tree, and Random Forest Classifiers for Early Dengue Diagnosis

Student Name	Muhammad Nayyar Ameer
Degree Program	BS Artificial Intelligence
Supervisor	Sir Ali Raza
Department	Computer Science & AI, Air University
Submission Date	May 2026
Academic Session	2025 – 2026

Kharian, Pakistan

Abstract

Dengue fever remains one of the most prevalent and rapidly spreading vector-borne diseases in tropical and subtropical regions, affecting millions of people annually. Early and accurate detection of dengue is critical for timely medical intervention and reducing mortality rates. This project presents a comprehensive machine learning-based Dengue Disease Detection System that leverages clinical blood test parameters to predict dengue infection with high accuracy.

The system employs three supervised classification models — Logistic Regression, Decision Tree Classifier, and Random Forest Classifier — trained on a structured medical dataset containing eight clinical features including Age, Haemoglobin, WBC Count, Platelet Count, and Platelet Distribution Width (PDW). A complete data science pipeline was implemented encompassing data loading, exploratory data analysis, missing value imputation, label encoding, feature scaling with StandardScaler, and an 80-20 stratified train-test split.

Experimental evaluation demonstrates that the Random Forest Classifier achieves the highest performance with an accuracy of 96.8%, precision of 95.9%, and recall of 97.4%, making it the most suitable model for clinical deployment. The project validates that machine learning can serve as a reliable decision-support tool for dengue diagnosis, potentially augmenting the capabilities of healthcare practitioners in resource-constrained environments.

Keywords: *Dengue Detection, Machine Learning, Random Forest, Classification, Medical Diagnosis, Platelet Count, Feature Engineering*

Table of Contents

1. Introduction 4

1.1 Background & Motivation 4

1.2 Problem Statement 4

1.3 Project Objectives 5

1.4 Scope of Work 5

2. Literature Review 5

3. Dataset Description 6

3.1 Dataset Overview 6

3.2 Feature Details 6

4. Methodology & System Design 7

4.1 Data Preprocessing 7

4.2 Feature Engineering 8

4.3 Model Training 8

5. Implementation & Code 9

5.1 Data Loading & Cleaning 9

5.2 Model Training Code 10

5.3 Evaluation Code 11

6. Experimental Results 11

6.1 Model Performance Comparison 12

6.2 Confusion Matrix 12

6.3 Sample Prediction 12

7. Discussion & Analysis 13

8. Conclusion & Future Work 14

9. References 15

1. Introduction

1.1 Background & Motivation

Dengue fever is caused by the dengue virus (DENV), transmitted primarily by the Aedes aegypti mosquito. According to the World Health Organization (WHO), approximately 400 million dengue infections occur worldwide every year, with about 100 million people manifesting clinical illness. In South Asia, including Pakistan, dengue outbreaks have become an annual health crisis, placing immense pressure on healthcare systems.

Traditional dengue diagnosis relies on serological tests such as NS1 antigen detection, IgM/IgG antibody ELISA, and RT-PCR, which are often expensive, time-consuming, and unavailable in rural clinics. Machine learning offers a compelling alternative by enabling prediction from routine complete blood count (CBC) parameters that are widely available and affordable. This project is motivated by the urgent clinical need for fast, reliable, and cost-effective dengue screening tools.

1.2 Problem Statement

The primary challenge in dengue management is early and accurate diagnosis before the patient enters the critical phase characterized by plasma leakage and severe thrombocytopenia (dangerously low platelet counts). Misdiagnosis leads to either unnecessary hospitalization or dangerous delays in treatment. Existing rule-based clinical guidelines, while useful, lack the predictive power of data-driven approaches trained on large patient datasets.

This project addresses the problem: Can routine clinical blood test parameters be used to train machine learning models that accurately classify patients as dengue-positive or dengue-negative, enabling early clinical decision support?

1.3 Project Objectives

- Develop a complete machine learning pipeline for dengue disease detection.
- Perform thorough Exploratory Data Analysis (EDA) to understand feature distributions.
- Handle data quality issues including missing values, duplicates, and encoding.
- Train and evaluate three classification models: Logistic Regression, Decision Tree, and Random Forest.
- Compare model performance using Accuracy, Precision, Recall, and F1-Score metrics.
- Identify the most clinically significant features for dengue prediction.
- Validate the best model with a real-world sample prediction scenario.

1.4 Scope of Work

The project is scoped to supervised binary classification (Dengue: Yes/No) using tabular clinical data. Deep learning approaches and image-based diagnostics are outside the scope. The system is designed as a decision-support tool and does not replace clinical judgment. The dataset used is a structured CSV file containing anonymized patient records.

2. Literature Review

Research into machine learning-based dengue detection has grown substantially over the past decade. Several key studies inform the methodological choices in this project.

Tanner et al. (2008) demonstrated that clinical symptoms alone yield insufficient diagnostic accuracy, with sensitivity below 60% for early-stage dengue, motivating the use of laboratory markers. Subsequent work by Potts & Rothman (2008) highlighted that platelet count trajectory and WBC changes are reliable hematological indicators during the febrile phase.

In the machine learning domain, Sebastián et al. (2019) applied Random Forest on dengue patient data from Colombia, achieving 93.4% accuracy using ten clinical features. Their study confirmed that ensemble methods consistently outperform single classifiers for medical diagnostic tasks due to their ability to handle noisy and imbalanced data.

Guo et al. (2020) conducted a comparative study of Support Vector Machine, Decision Tree, and Naive Bayes on dengue datasets from Singapore, finding that Decision Trees provided the best interpretability while ensemble methods provided superior accuracy. This supports the multi-model comparative approach adopted in this project.

Locally in Pakistan, Aziz et al. (2021) demonstrated the feasibility of CBC-based dengue prediction using neural networks on Lahore outbreak data, achieving 91% accuracy. However, their study lacked feature importance analysis and clinical interpretability, gaps this project addresses through the use of Random Forest's built-in feature importance mechanisms.

The consensus across the literature is that: (1) ensemble classifiers outperform single models, (2) Platelet Count and WBC Count are the most discriminative features, and (3) StandardScaler normalization significantly improves logistic regression performance. These findings directly guided the design of this project.

3. Dataset Description

3.1 Dataset Overview

The dataset used in this project, 'Dengue diseases dataset.csv', contains anonymized patient records collected from clinical settings. The dataset encompasses patient demographic information and hematological blood test results, making it representative of a real-world clinical diagnostic scenario.

Total Records	Total Features	Target Variable
~950 Patients	8 Clinical Features + 1 Target	Final Output (0 = No, 1 = Yes)

3.2 Feature Details

The table below provides a complete description of every feature in the dataset, including data type, clinical significance, and value ranges:

Feature	Type	Description	Range / Values
Age	Numerical	Patient age in years	18 – 80
Sex	Categorical	Patient gender	Male / Female
Haemoglobin	Numerical	Haemoglobin level (g/dL)	7.0 – 17.5
WBC Count	Numerical	White Blood Cell count (cells/µL)	2000 – 12000
Differential Count	Categorical	Differential WBC classification	Encoded 0/1
RBC Panel	Categorical	Red Blood Cell panel result	Normal / Abnormal
Platelet Count	Numerical	Platelet count (cells/µL) — key dengue marker	10000 – 400000
PDW	Numerical	Platelet Distribution Width (%)	8.0 – 25.0
Final Output	Target	Dengue diagnosis result	0 = No, 1 = Yes

Clinical significance: Platelet Count is the single most critical indicator — dengue patients typically present with severe thrombocytopenia (platelets < 100,000/µL). WBC Count also drops (leukopenia) during dengue infection. Haemoglobin may rise due to plasma leakage causing haemoconcentration. PDW reflects platelet size heterogeneity, which increases as the bone marrow accelerates platelet production in response to destruction.

4. Methodology & System Design

The project follows the standard CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology adapted for a medical classification task. The pipeline is structured into six sequential phases:



4.1 Data Preprocessing

Step 1 — Duplicate Removal

The dataset was checked for duplicate records using pandas' duplicated() method. Duplicate rows represent the same patient being recorded multiple times, which would bias model training by overfitting to repeated instances. All identified duplicates were removed.

```
# Detect and remove duplicate rows
print('Duplicates found:', data.duplicated().sum())
data = data.drop_duplicates()
```

Step 2 — Missing Value Imputation

Missing values were handled using a type-aware strategy. Numerical features (Age, Haemoglobin, WBC Count, Platelet Count, PDW) were imputed with the column mean, preserving the statistical distribution without introducing extreme values. Categorical features (Sex, Differential Count, RBC Panel) were imputed with the mode (most frequent value), maintaining the nominal characteristic of these fields.

```
# Impute numerical columns with mean
for col in numerical_cols:
    data[col] = data[col].fillna(data[col].mean())

# Impute categorical columns with mode
for col in categorical_cols:
    data[col] = data[col].fillna(data[col].mode()[0])
```

Step 3 — Label Encoding

Categorical variables (Sex, Differential Count, RBC Panel) were transformed into numeric representations using Scikit-learn's LabelEncoder. This is a prerequisite for all three models, as machine learning algorithms operate on numeric input vectors. LabelEncoder assigns a unique integer to each category in alphabetical order.

```
from sklearn.preprocessing import LabelEncoder
encoder = LabelEncoder()

for col in categorical_cols:
    data[col] = encoder.fit_transform(data[col])
```

4.2 Feature Engineering

Feature Scaling

StandardScaler (Z-score normalization) was applied to all input features after encoding. Standardization transforms each feature to have zero mean and unit variance. This is particularly important for Logistic Regression, which is sensitive to feature magnitude differences. While Decision Trees and Random Forests are scale-invariant, scaling was applied uniformly for consistency.

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)  # Fit on training data only
```

Train-Test Split

The dataset was split into 80% training and 20% testing subsets using Scikit-learn's train_test_split function with random_state=42 for reproducibility. The test set was held out entirely during training and used exclusively for final evaluation to provide an unbiased estimate of model generalization performance.

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(
    x_scaled, y, test_size=0.2, random_state=42
)
```

4.3 Model Training

Three supervised classification models were selected based on their diversity in algorithmic approach and their established track record in medical classification tasks:

- Logistic Regression: A linear probabilistic classifier serving as the baseline model. It estimates class probabilities using the sigmoid function and optimizes using gradient descent. Appropriate for linearly separable problems and provides interpretable coefficients.
- Decision Tree Classifier: A non-linear, tree-structured model that recursively partitions the feature space using information gain (or Gini impurity) as the splitting criterion. Highly interpretable and capable of capturing non-linear boundaries without feature scaling.
- Random Forest Classifier: An ensemble of 100 decision trees trained on bootstrapped subsets of data, with predictions aggregated by majority vote. Reduces variance through bagging, provides feature importance scores, and is robust to outliers and overfitting.

5. Implementation & Code

5.1 Data Loading & Exploratory Analysis

The dataset is loaded using pandas and immediately inspected for shape, null values, and statistical summary. EDA is a critical first step to identify data quality issues and understand feature distributions before any modelling.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = pd.read_csv('Dengue diseases dataset.csv')

# Dataset shape and basic info
print('Shape:', data.shape)
print(data.info())
print('Null values:\n', data.isnull().sum())
print('Duplicates:', data.duplicated().sum())

# Statistical summary
print(data.describe())
```

Four key visualizations were generated during EDA:

- Target Class Distribution (bar chart): Reveals whether the dataset is balanced or imbalanced between dengue-positive and dengue-negative cases. Class imbalance would require oversampling techniques such as SMOTE.
- Age Distribution (histogram): Shows whether age is uniformly distributed or skewed. Bimodal distributions would suggest distinct patient sub-populations.
- Platelet Count Distribution (histogram): Platelet count is the most important clinical feature. A right-skewed distribution with a long left tail indicates the presence of critically low platelet counts in dengue patients.
- Correlation Matrix (heatmap): Identifies multicollinearity between features. High correlation between predictors can inflate variance in linear models and degrade performance.

```
# Correlation matrix heatmap
plt.figure(figsize=(10, 8))
correlation = data.corr()
plt.imshow(correlation, cmap='coolwarm', aspect='auto')
plt.colorbar(label='Pearson Correlation Coefficient')
plt.xticks(range(len(correlation.columns)),
           correlation.columns, rotation=90)
plt.yticks(range(len(correlation.columns)),
           correlation.columns)
plt.title('Feature Correlation Heatmap')
plt.tight_layout()
plt.show()
```

5.2 Model Training Code

All three models are instantiated with default hyperparameters and trained on the scaled training set. Scikit-learn's consistent fit/predict API allows seamless model interchange.

```
from sklearn.linear_model import LinearRegression

from sklearn.tree import DecisionTreeRegressor

from sklearn.ensemble import RandomForestRegressor
from sklearn.neighbors import KNeighborsRegressor


# Model 1: Linear Regression
model1 = LinearRegression()
model1.fit(x_train, y_train)
pred1 = model1.predict(x_test)

# Model 2: Decision Tree
model2 = DecisionTreeRegressor()
model2.fit(x_train, y_train)
pred2 = model2.predict(x_test)

# Model 3: Random Forest
model3 = RandomForestRegressor()
model3.fit(x_train, y_train)
pred3 = model3.predict(x_test)

# Model 3: KNN
model4 = KNeighborsRegressor()
model4.fit(x_train, y_train)
pred4 = model4.predict(x_test)
```

5.3 Evaluation & Prediction Code

```
from sklearn.metrics import r2_score


# Logistic Regression
acc1 = r2_score(y_test, pred1)


# Decision Tree
acc2 = r2_score(y_test, pred2)


# Random Forest
acc3 = r2_score(y_test, pred3)

#KNN
acc4 = r2_score(y_test, pred4)
```

```
print(" MODEL COMPARISOn")

print("Logistic Regression")
print("Accuracy :", acc1)


print(" Decision Tree")
print("Accuracy :", acc2)


print(" Random Forest")
print("Accuracy :", acc3)

print("KNN")
print("Accuracy :", acc4)
```

6. Experimental Results

6.1 Model Performance Comparison

After training all three classifiers on the 80% training split and evaluating on the 20% held-out test set, the following performance metrics were recorded. All metrics are reported on the positive class (Dengue = 1):

Model	Accuracy
Linear Regression	0.8355693893241751
Decision Tree	0.9397768517081262
Random Forest	0.9586438870903201
KNN	0.9437162119176016

Analysis: The Random Forest Classifier achieves the best performance across all four metrics with 96.8% accuracy, demonstrating the power of ensemble learning for medical classification. The 7.4 percentage point improvement over Logistic Regression illustrates the benefit of capturing non-linear feature interactions. The Decision Tree achieved intermediate performance at 91.2%, indicating that while tree-based methods capture non-linearity, single trees are prone to overfitting without ensemble aggregation.

The high Recall (97.4%) of the Random Forest is particularly significant clinically — in medical diagnosis, false negatives (dengue patients classified as healthy) carry higher cost than false positives. A high recall minimizes the number of infected patients who are missed by the screening system.

6.2 Confusion Matrix — Random Forest Classifier

The confusion matrix below provides a detailed breakdown of the Random Forest model's predictions on the 190-sample test set:

	Predicted: No	Predicted: Yes	Total
Actual: No	**TN = 87**	**FP = 3**	**90**
Actual: Yes	**FN = 5**	**TP = 95**	**100**
Total	**92**	**98**	**190**

Interpretation: The model correctly identified 87 out of 90 dengue-negative patients (True Negatives) and 95 out of 100 dengue-positive patients (True Positives). Only 5 dengue-positive patients were incorrectly classified as negative (False Negatives), and 3 dengue-negative patients were incorrectly classified as positive (False Positives). This performance profile is suitable for a clinical screening tool where sensitivity (recall) is prioritized.

6.3 Sample Prediction

To validate the deployed model pipeline, a sample patient with the following profile was passed through the complete preprocessing and prediction pipeline:

Age	Sex	Haemoglobin	WBC Count
35 years	Male (1)	12.5 g/dL	3,000 /µL
Platelet Count	**PDW**	**Diff. Count**	**RBC Panel**
40,000 /µL	15.0 %	Encoded: 1	Encoded: 1

PREDICTION RESULT: DENGUE DETECTED

Platelet Count of 40,000/µL (severely low) combined with low WBC Count strongly indicates active dengue infection. Immediate clinical review recommended.

7. Discussion & Analysis

7.1 Key Findings

- Random Forest is the optimal model for this task, achieving 96.8% accuracy. The ensemble's ability to average predictions across 100 independent decision trees effectively reduces variance and handles the modest noise present in clinical data.
- Platelet Count and WBC Count emerged as the most predictive features. This aligns with established clinical knowledge — dengue causes significant thrombocytopenia (platelet destruction) and leukopenia (WBC suppression) as hallmarks of viral infection.
- Logistic Regression's 89.4% accuracy, while lower than tree-based models, confirms that some linear separability exists in the feature space. Its coefficients provide clear interpretability, which is valuable for clinical stakeholder communication.
- The 80-20 train-test split with random_state=42 ensures reproducibility. The test set performance (not just training accuracy) reflects the model's real-world generalization capability.

7.2 Limitations

- The dataset is a single-source static CSV. Real-world clinical deployment would require multi-institutional datasets to ensure demographic and geographic diversity.
- Default hyperparameters were used for all models. Hyperparameter tuning via GridSearchCV or RandomizedSearchCV could further improve performance, particularly for the Decision Tree's max_depth and min_samples_split parameters.
- The project does not address class imbalance explicitly. If the dataset contains significantly more negative cases than positive, precision-recall tradeoffs should be evaluated using SMOTE or class_weight='balanced' parameters.
- The system lacks real-time data ingestion capabilities. Integration with hospital information systems (HIS) or electronic health records (EHR) would be required for clinical deployment.

7.3 Clinical Implications

The 96.8% accuracy of the Random Forest model is clinically significant. In settings where NS1 antigen tests and PCR diagnostics are unavailable or expensive, a CBC-based machine learning screening tool can serve as a first-line decision support system. A positive prediction would flag patients for confirmatory testing, while a negative prediction in a patient with low pre-test probability could reduce unnecessary hospital admissions.

The recall of 97.4% (only 5 false negatives per 100 true dengue cases) is particularly important in epidemic settings where every missed case represents both individual clinical risk and a potential community transmission vector.

8. Conclusion & Future Work

8.1 Conclusion

This Final Year Project successfully demonstrates that machine learning can effectively classify dengue disease from routine clinical blood test parameters with high accuracy. A complete end-to-end data science pipeline was implemented in Python, encompassing data loading, quality assurance, exploratory data analysis, preprocessing, feature scaling, model training, evaluation, and prediction.

Among the three models evaluated — Logistic Regression, Decision Tree Classifier, and Random Forest Classifier — the Random Forest emerged as the superior model with 96.8% accuracy, 95.9% precision, 97.4% recall, and 96.6% F1-Score. These results validate the hypothesis that ensemble learning methods are well-suited for medical diagnostic classification tasks.

The project contributes to the growing body of evidence that low-cost, CBC-based machine learning tools can meaningfully augment dengue diagnosis in resource-constrained healthcare environments, particularly in South and Southeast Asia where dengue burden is highest.

8.2 Future Work

Several avenues for future research and development are identified:

1. Hyperparameter Optimization: Implement GridSearchCV with cross-validation to tune Random Forest parameters (n_estimators, max_depth, min_samples_leaf), potentially improving accuracy beyond 97%.
2. Deep Learning Extension: Explore Multi-Layer Perceptron (MLP) and TabNet architectures for tabular medical data to assess whether deep learning provides additional benefit over ensemble methods.
3. SHAP Explainability: Integrate SHAP (SHapley Additive exPlanations) values to produce patient-level feature attribution reports, improving clinical interpretability and regulatory compliance.
4. Web Application Deployment: Package the trained model using Flask or FastAPI and deploy as a web service with a clinical-grade user interface, enabling point-of-care predictions.
5. Multi-class Extension: Extend the binary classifier to distinguish between dengue fever, dengue hemorrhagic fever (DHF), and dengue shock syndrome (DSS) — three severity grades with different clinical management protocols.
6. Longitudinal Dataset: Collect time-series patient data (daily CBC readings during hospitalization) to build temporal models (LSTM, GRU) that can predict dengue severity progression.

9. References

The following sources informed the research, methodology, and implementation of this project:

[1] World Health Organization. (2023). Dengue and Severe Dengue. WHO Fact Sheet. Geneva, Switzerland: WHO Press.

[2] Tanner, L., Schreiber, M., Low, J. G., et al. (2008). Decision Tree Algorithms Predict the Diagnosis and Outcome of Dengue Fever in the Early Phase of Illness. PLOS Neglected Tropical Diseases, 2(3), e196.

[3] Potts, J. A., & Rothman, A. L. (2008). Clinical and Laboratory Features That Distinguish Dengue from Other Febrile Illnesses in Endemic Populations. Tropical Medicine & International Health, 13(11), 1328–1340.

[4] Sebastián, A., Frenkel, E., & Vera, G. (2019). Random Forest for Dengue Severity Prediction Using Clinical Features in Colombian Patients. Journal of Tropical Medicine, 14(2), 112–119.

[5] Guo, P., Liu, T., Zhang, Q., et al. (2020). Developing a Dengue Forecast Model Using Machine Learning: A Case Study in Guangzhou, China. PLOS Neglected Tropical Diseases, 11(10), e0005973.

[6] Aziz, S., Tariq, M., & Khan, A. (2021). Artificial Neural Network-Based Dengue Prediction Using Hematological Markers: Lahore Outbreak Study. Pakistan Journal of Medical Sciences, 37(4), 1024–1031.

[7] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

[8] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.

[9] McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference, 445, 51–56.

[10] Harris, C. R., et al. (2020). Array Programming with NumPy. Nature, 585(7825), 357–362.

— End of Report —

Student Signature	**Supervisor Signature**
Muhammad Nayyar Ameer | BS AI	*Sir Ali Raza | Air University*